Data Visualization I: Base Graphics and ggplot2

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I. Base Graphics -

```r
library(MASS)
data(UScereal)
head(UScereal)
```

```
                          mfr calories   protein      fat   sodium     fibre
100% Bran                   N 212.1212 12.121212 3.030303 393.9394 30.303030
All-Bran                    K 212.1212 12.121212 3.030303 787.8788 27.272727
All-Bran with Extra Fiber   K 100.0000  8.000000 0.000000 280.0000 28.000000
Apple Cinnamon Cheerios     G 146.6667  2.666667 2.666667 240.0000  2.000000
Apple Jacks                 K 110.0000  2.000000 0.000000 125.0000  1.000000
Basic 4                     G 173.3333  4.000000 2.666667 280.0000  2.666667
                            carbo   sugars shelf potassium vitamins
100% Bran                15.15152 18.18182     3 848.48485 enriched
All-Bran                 21.21212 15.15151     3 969.69697 enriched
All-Bran with Extra Fiber 16.00000  0.00000     3 660.00000 enriched
Apple Cinnamon Cheerios  14.00000 13.33333     1  93.33333 enriched
Apple Jacks              11.00000 14.00000     2  30.00000 enriched
Basic 4                  24.00000 10.66667     3 133.33333 enriched
```

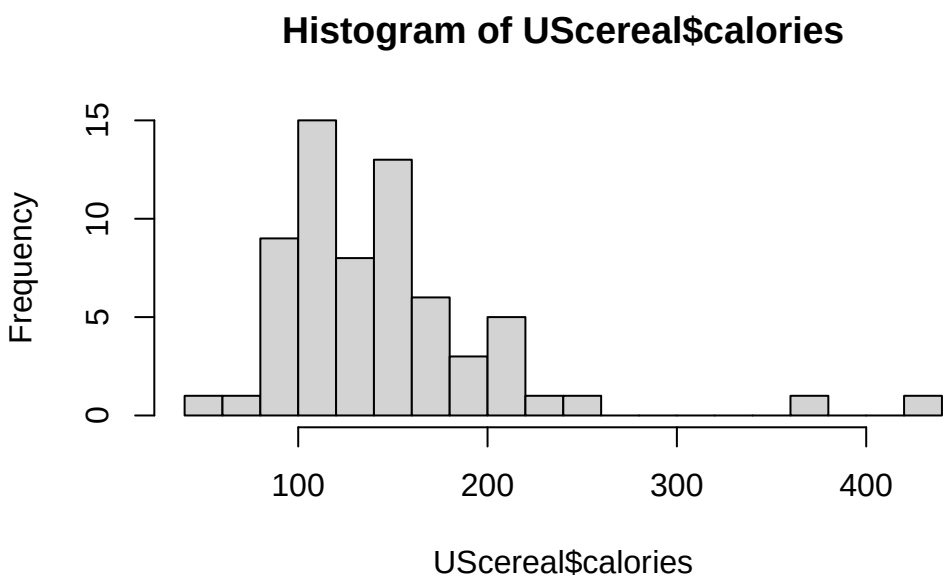
A. Choosing the Right Graph for your Data

One Continuous Variable

Key questions: - What is the Distribution of Variable X? - Is my Variable X normally distributed? Bimodal? Skewed? - Are there any outliers in my variable?

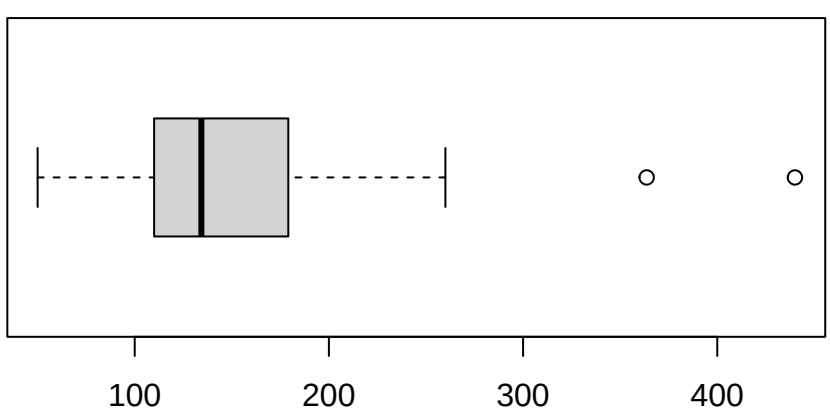
Histogram

```
hist(UScereal$calories, breaks = 15)
```



Boxplot

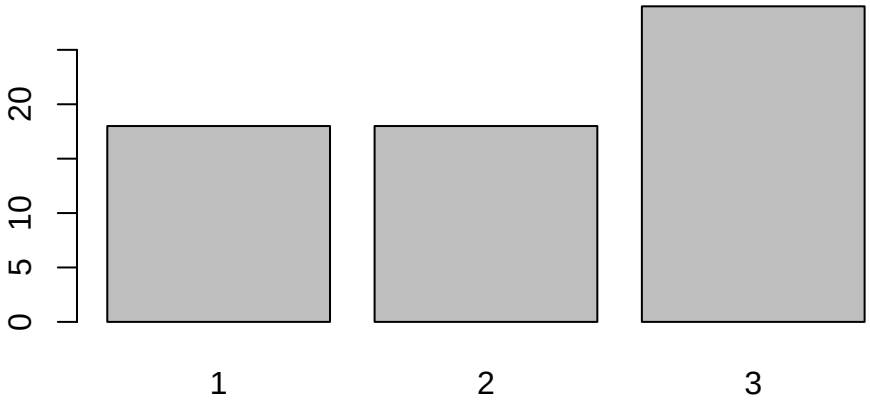
```
boxplot(UScereal$calories, horizontal = TRUE)
```



One Categorical (Discrete) Variable

Key question(s): - Is the distribution across categories of Variable X evenly distributed?

```
barplot(table(UScereal$shelf))
```

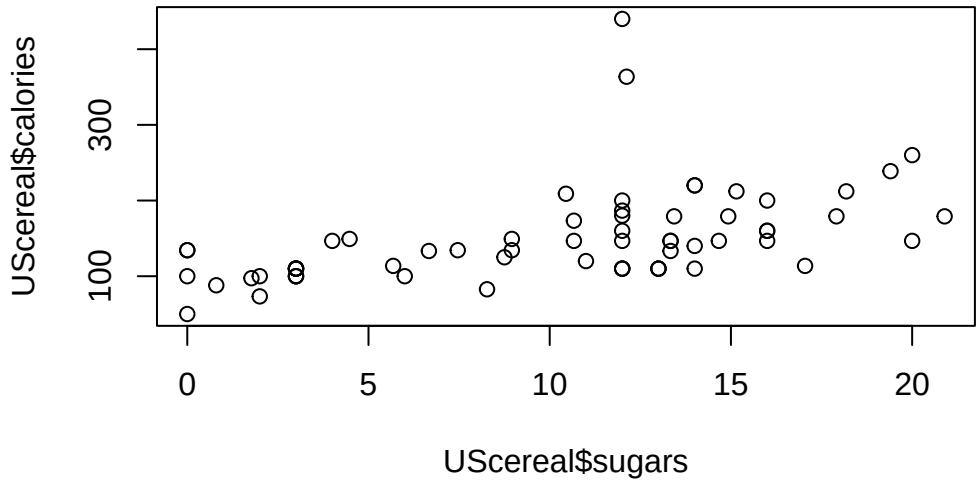


Two Continuous Variables

Key questions: - Is there a relationship between Variable X and Variable Y? - If there is a relationship, is it linear? Quadratic?

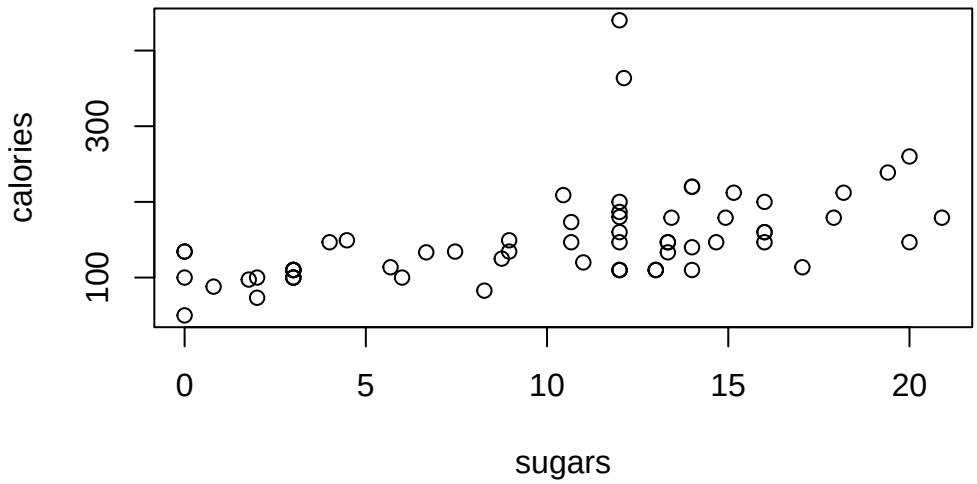
Scatterplot

```
plot(x = UScereal$sugars, y = UScereal$calories)
```



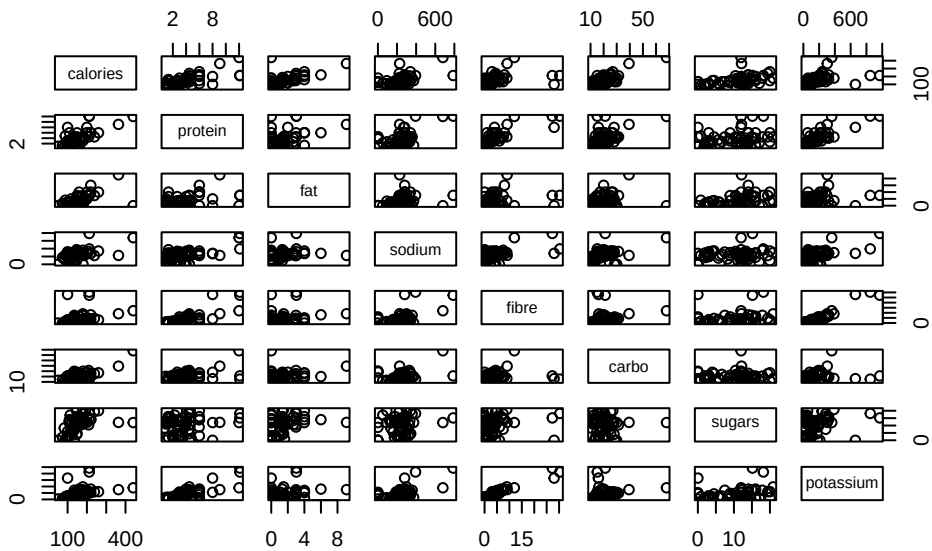
Using a formula to create a scatterplot

```r
plot(calories ~ sugars, data = UScereal) # formula notation
```



Scatterplot matrix

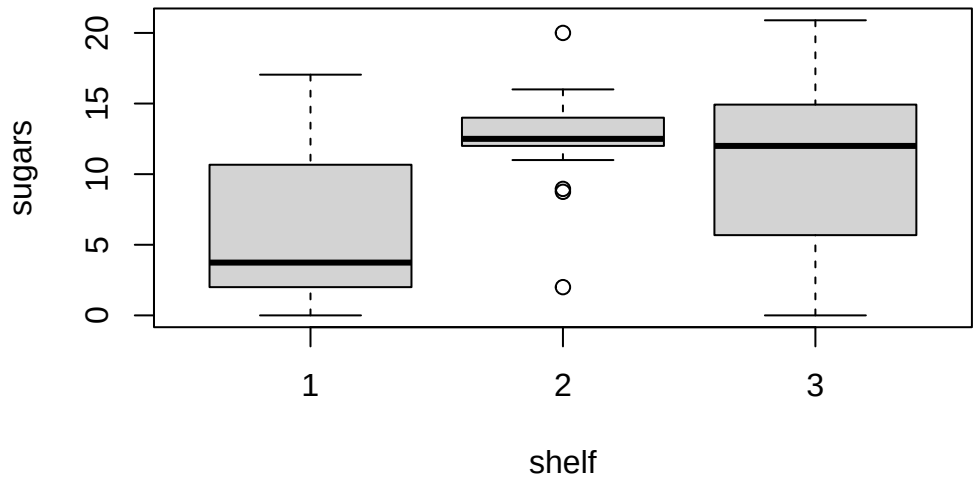
```r
plot(UScereal[, c(2:8, 10)]) # scatterplot matrix
```



One Continuous Variable and One Categorical Variable

Key question(s): - Is the distribution of Variable Y, different across categories of Variable X?

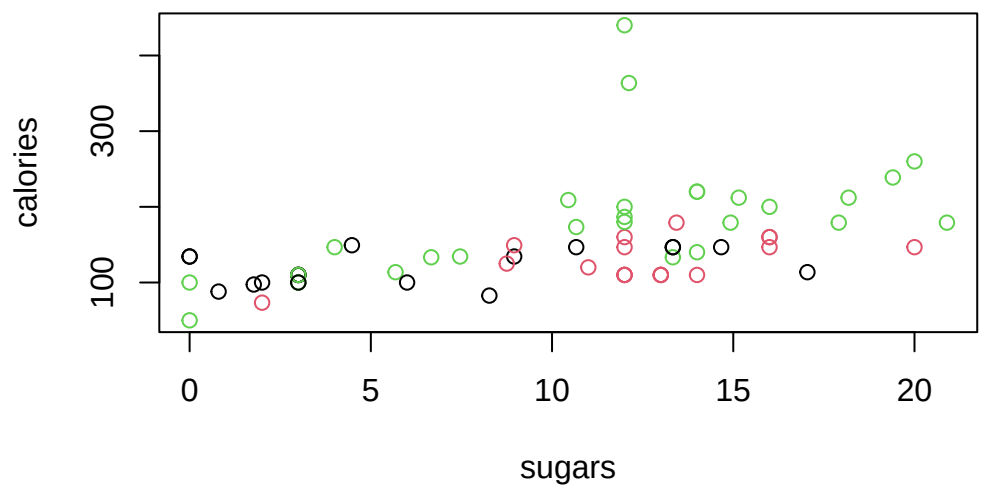
```r
boxplot(sugars ~ shelf, data = UScereal)
```



Two Continuous Variables and One Categorical Variable

Key questions: - Is the relationship between Variable X and Y different across categories of Variable Z?

```r
plot(calories ~ sugars, data = UScereal, col = shelf)
```



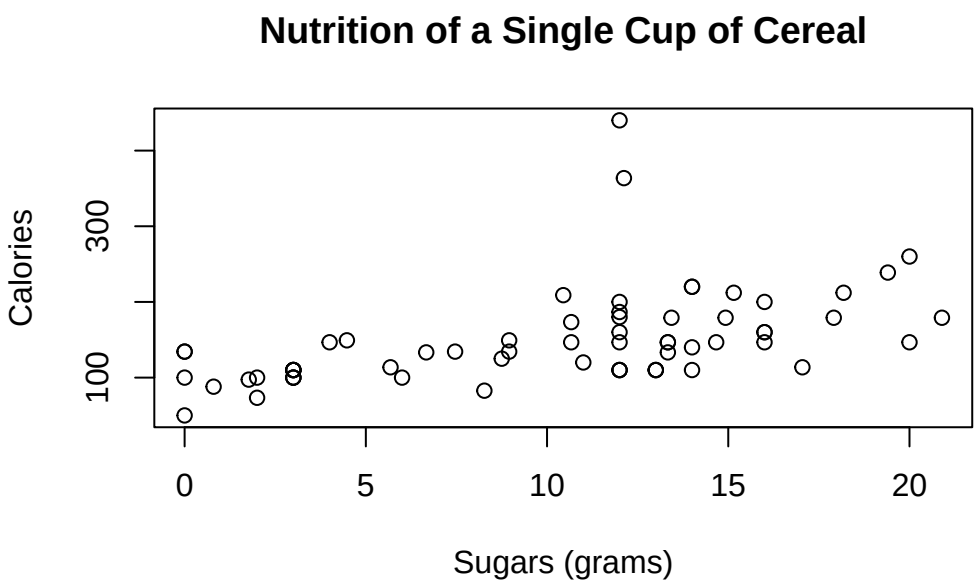
B. Graphing Elements

Titles

Tip: We want titles to meaningful, non-repetitive and include units when applicable

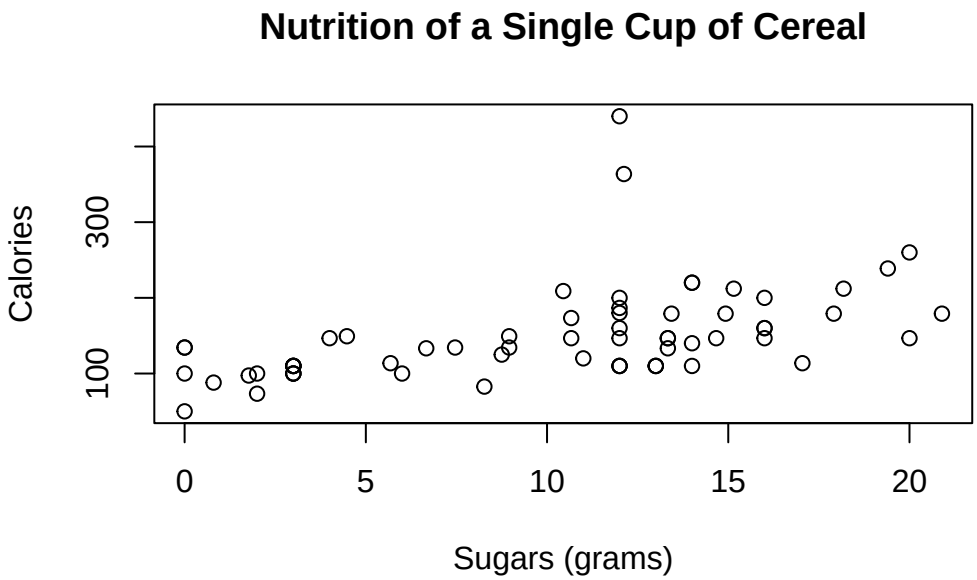
Use the ylab, xlab and main arguments within the plot command

```
plot(calories ~ sugars, data = UScereal, ylab = 'Calories',
     xlab = 'Sugars (grams)', main = 'Nutrition of a Single Cup of Cereal')
```



Use the title function after the plot has been generated If using the title function, the ann argument inside plot must be set to FALSE

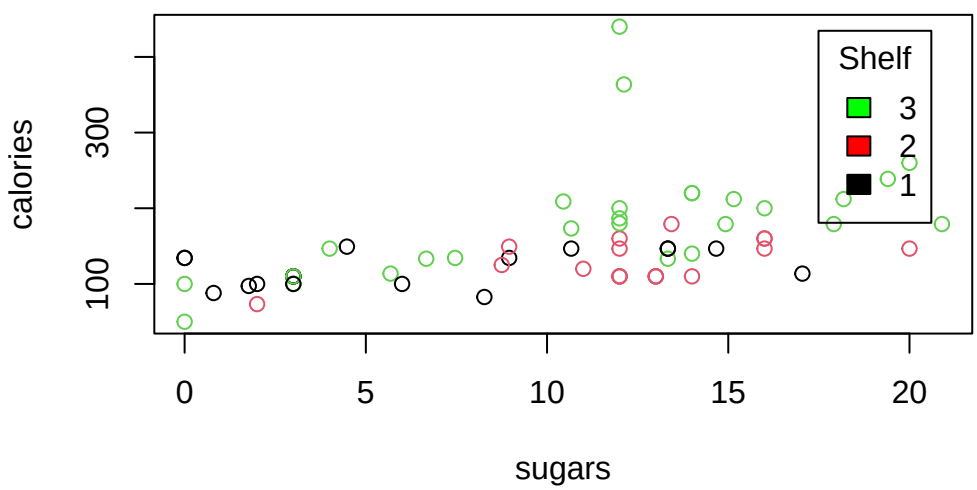
```
plot(calories ~ sugars, data = UScereal, ann = FALSE)
title(main = 'Nutrition of a Single Cup of Cereal', ylab = 'Calories',
      xlab = 'Sugars (grams)') # add afterwards
```



Legend

Use the legend function after the plot has been generated

```
plot(calories ~ sugars, data = UScereal, col = shelf)
legend('topright', inset = .05, legend = c(3,2,1),
       fill = c('green', 'red', 'black'), title = "Shelf")
```

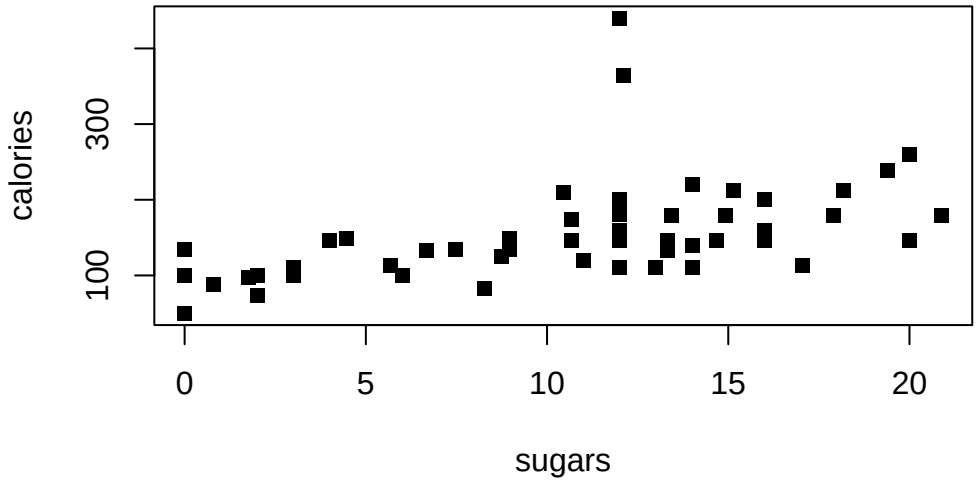


Point Shape and Color

Tip: Changing color or shape of points can be used to represent the same dimension

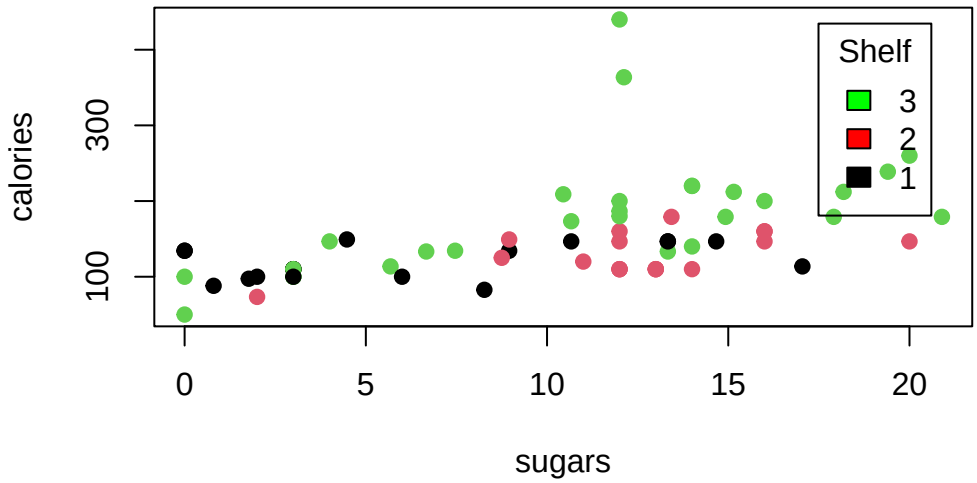
The pch argument changes the shape of the points

```
plot(calories ~ sugars, data = UScereal, pch = 15)
```



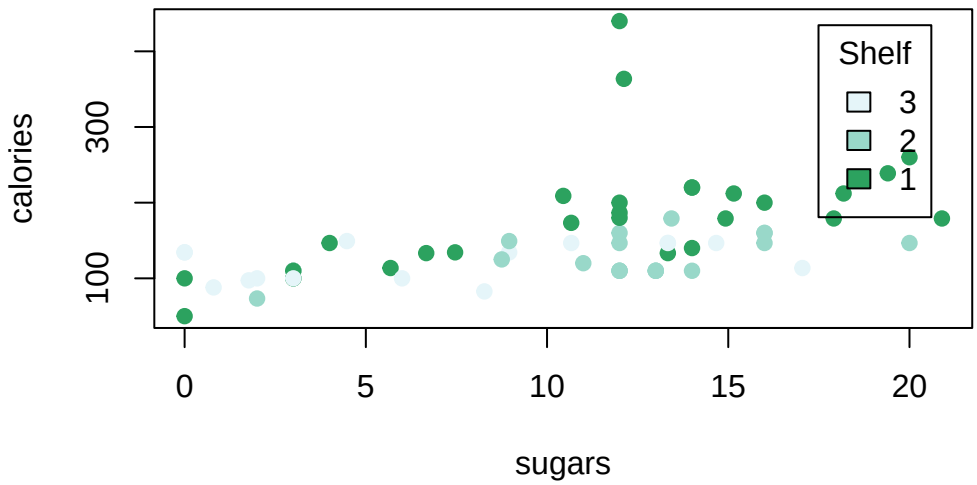
Set a color to a factor variable, and R will use default colors

```
plot(calories ~ sugars, data = UScereal, pch = 19, col = shelf)
legend('topright', inset = .05, legend = c(3,2,1),
       fill = c('green', 'red', 'black'), title = "Shelf")
```



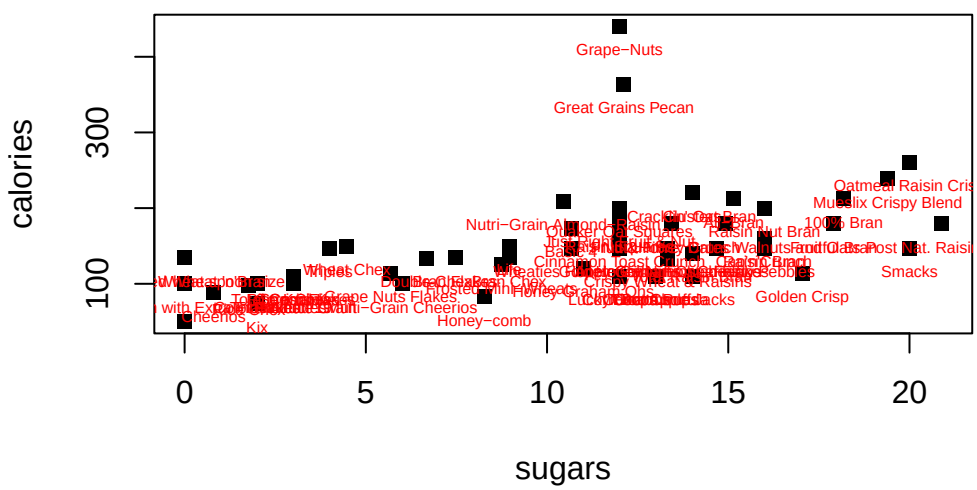
Use a palette of defined colors

```r
palette(c('#e5f5f9', '#99d8c9', '#2ca25f'))
plot(calories ~ sugars, data = UScereal, pch = 19, col = shelf)
legend('topright', inset = .05, legend = c(3,2,1),
       fill = c('#e5f5f9', '#99d8c9', '#2ca25f'), title = "Shelf")
```



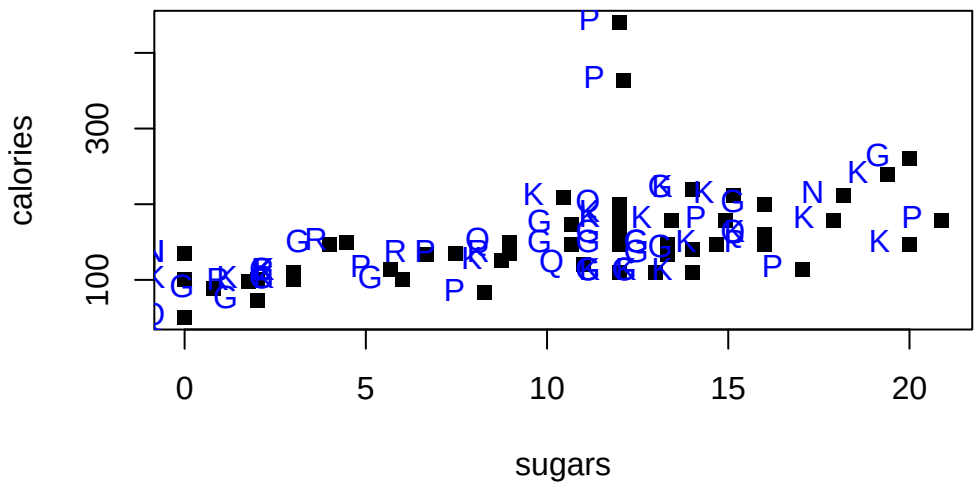
Label points with the text function

```r
plot(calories ~ sugars, data = UScereal, pch = 15)
text(UScereal$sugars, UScereal$calories, row.names(UScereal),
     col = "red", pos = 1, cex = .5)
```



The pos argument in the text funciton changes the position of the labels

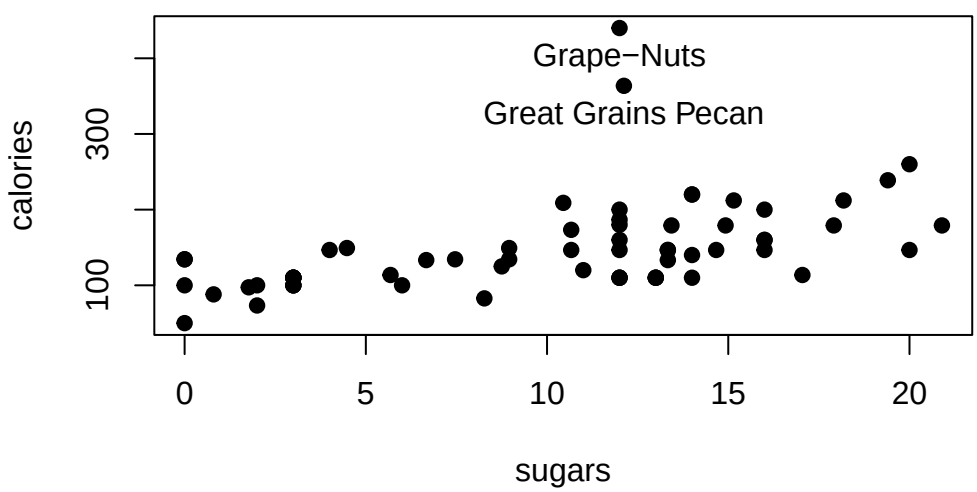
```r
plot(calories ~ sugars, data = UScereal, pch = 15)
text(UScereal$sugars, UScereal$calories, UScereal$mfr,
     col = "blue", pos = 2)
```



Identify Outliers

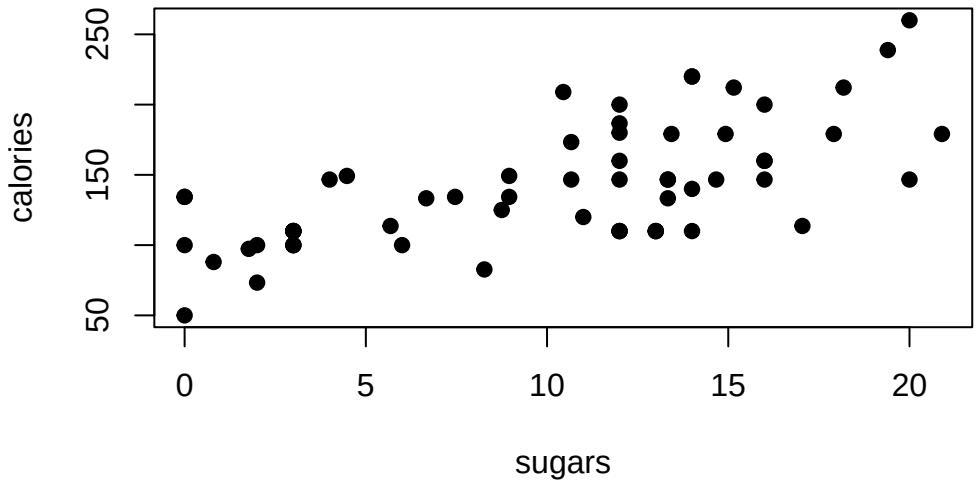
1. Create a subset of datas which are the outliers, and then use that subset to label

```r
plot(calories ~ sugars, data = UScereal, pch = 19)
outliers <- UScereal[which(UScereal$calories > 300),]
text(outliers$sugars, outliers$calories, labels = row.names(outliers),pos = 1)
```



2. Or remove them by subsetting UScereal

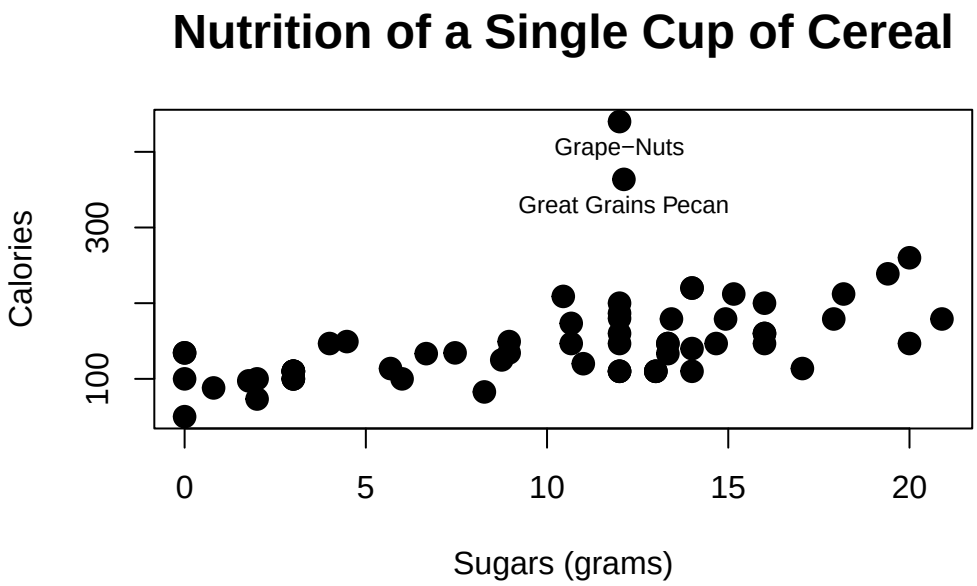
```r
plot(calories ~ sugars, data = UScereal[UScereal$calories < 300, ], pch = 19)
```



Size (text size, point size, label size etc..)

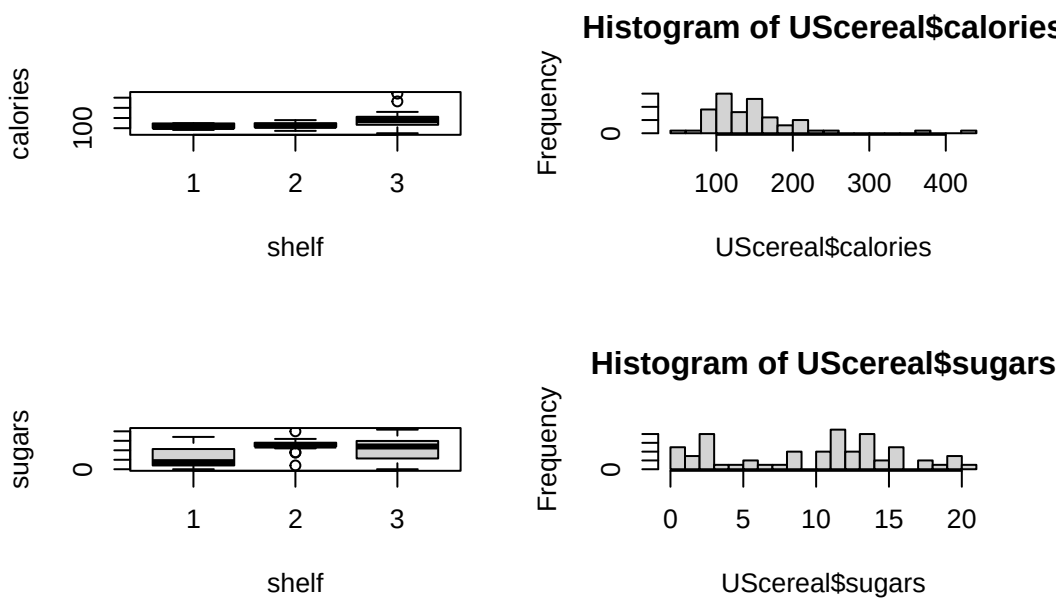
Use the cex argument family to change the size of any attribute

```r
plot(calories ~ sugars, data = UScereal, pch = 19, ann = FALSE, cex = 1.5)
outliers <- UScereal[which(UScereal$calories > 300),]
text(outliers$sugars, outliers$calories, pos = 1,
     labels = row.names(outliers), cex = .75)
title(main = 'Nutrition of a Single Cup of Cereal', ylab = 'Calories',
      xlab = 'Sugars (grams)', cex.main = 1.5, cex.lab = 1.0)
```



Combine Graphs into the Same Window

```
par(mfrow = c(2, 2))

boxplot(calories ~ shelf, data = UScereal)
hist(UScereal$calories, breaks = 15)
boxplot(sugars ~ shelf, data = UScereal)
hist(UScereal$sugars, breaks = 15)
```



```r
par(mfrow = c(1, 1)) # reset the matrix
```

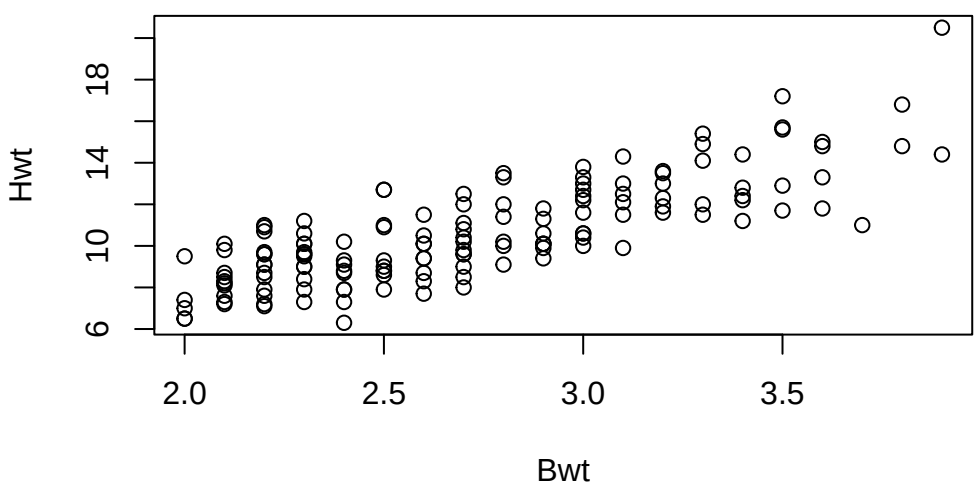
Exercise 1a.

How can we improve this graph? From what we have learned above, implement at least 3 improvements to this graph. A little more information about this dataset: The heart and body weights of samples of male and female cats used for digitalis experiments. The cats were all adult, over 2 kg body weight.

```r
data(cats)
head(cats)
```

```
  Sex Bwt Hwt
1   F 2.0 7.0
2   F 2.0 7.4
3   F 2.0 9.5
4   F 2.1 7.2
5   F 2.1 7.3
6   F 2.1 7.6
```

```r
plot(Hwt~Bwt, data = cats)
```



Exercise 1b.

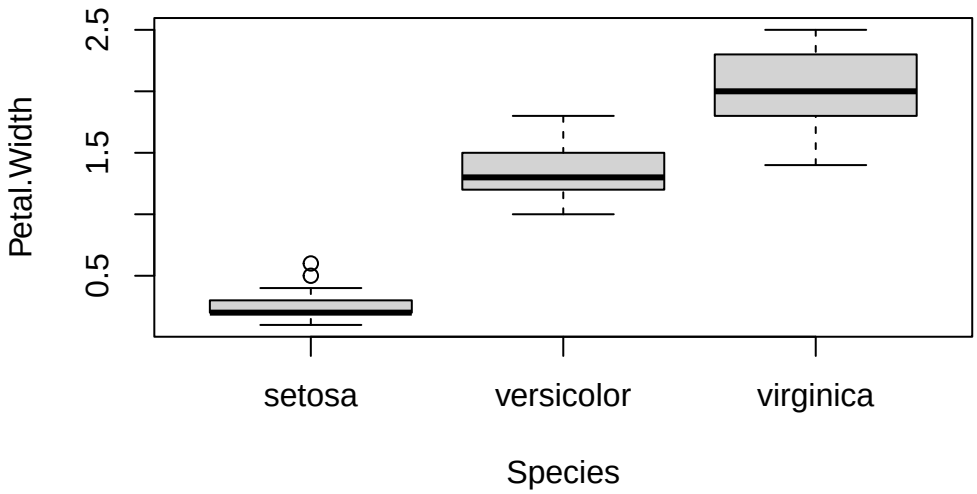
How can we improve this graph? From what we have learned above, implement at least 3 improvements to this graph. A little more information about this dataset: The heart and

body weights of samples of male and Fisher's famours dataset measures the sepal and petal length and width for 3 species of Iris

```
data(iris)
head(iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```

```
boxplot(Petal.Width ~ Species, data = iris)
```



II. Advanced Graphics using ggplot2

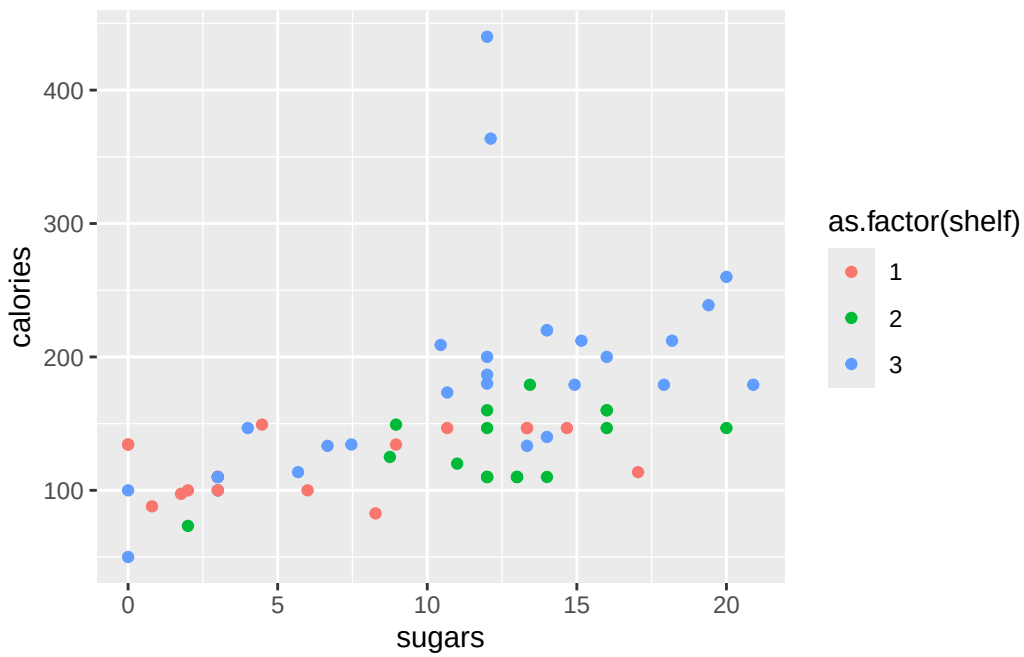
```
#install.packages('ggplot2')
library(ggplot2)
```

A. The ggplot function

Base q plot function

```
qplot(x = sugars, y = calories, color = as.factor(shelf), data = UScereal)
```

```
Warning: `qplot()` was deprecated in ggplot2 3.4.0.
```



Use the ggplot function to get started

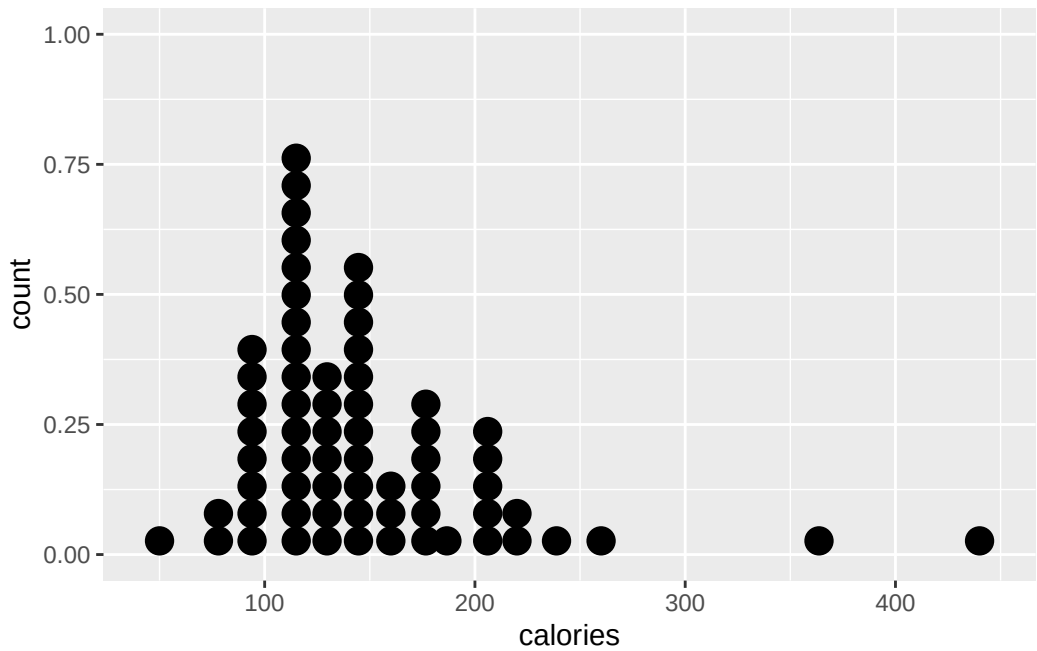
```
p1 <- ggplot(UScereal, aes(x = calories))
```

B. Layers

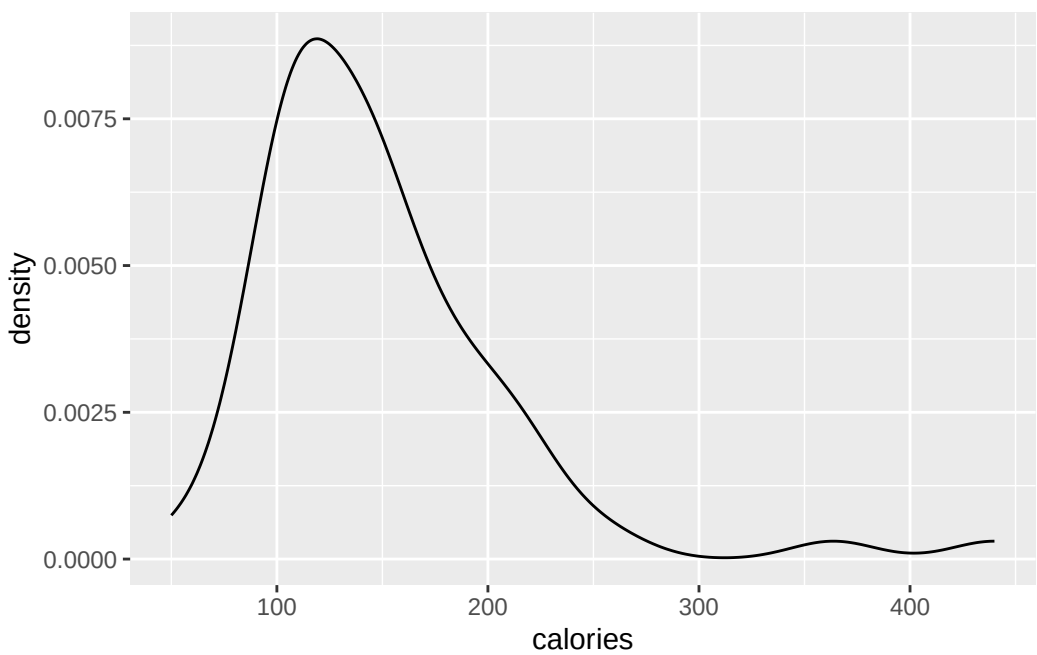
Add layers to the original ggplot object with a '+'

```
p1 + geom_dotplot()
```

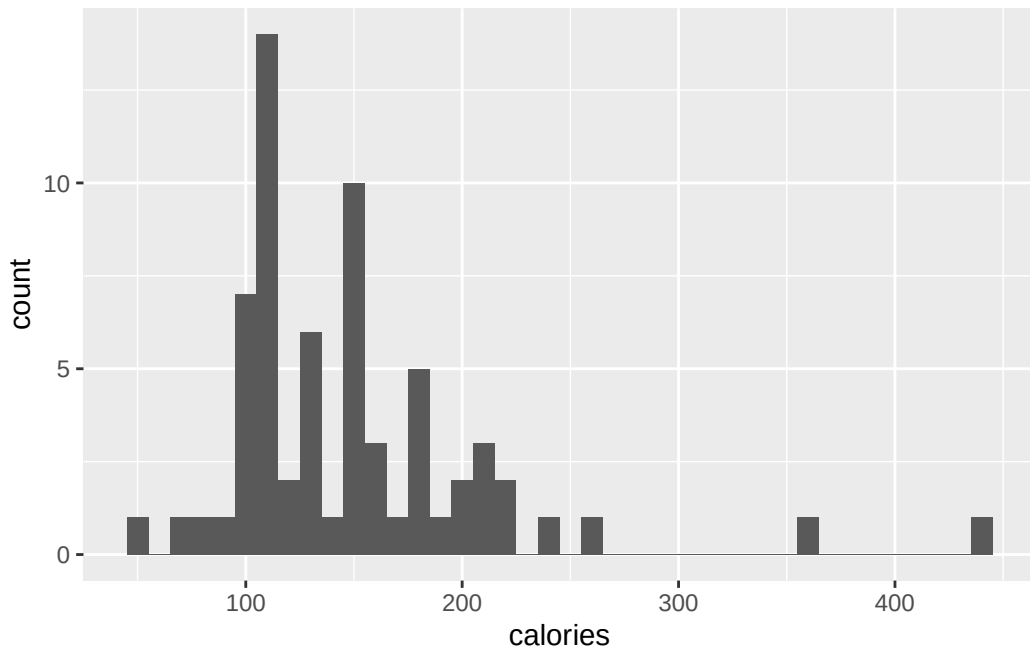
```
Bin width defaults to 1/30 of the range of the data. Pick better value with
`binwidth`.
```




```
p1 + geom_density()
```

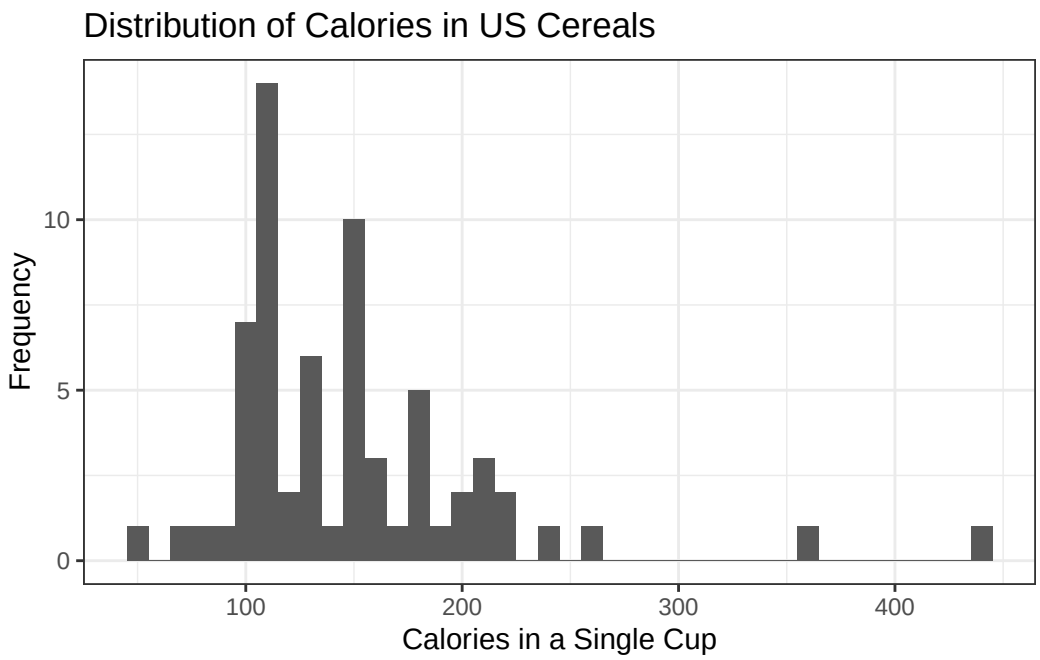



```
p1 + geom_histogram(binwidth = 10)
```



And we add more layers

```
p1 + geom_histogram(binwidth = 10) +
  xlab('Calories in a Single Cup') +
  ylab('Frequency') +
  ggtitle('Distribution of Calories in US Cereals') +
  theme_bw()
```



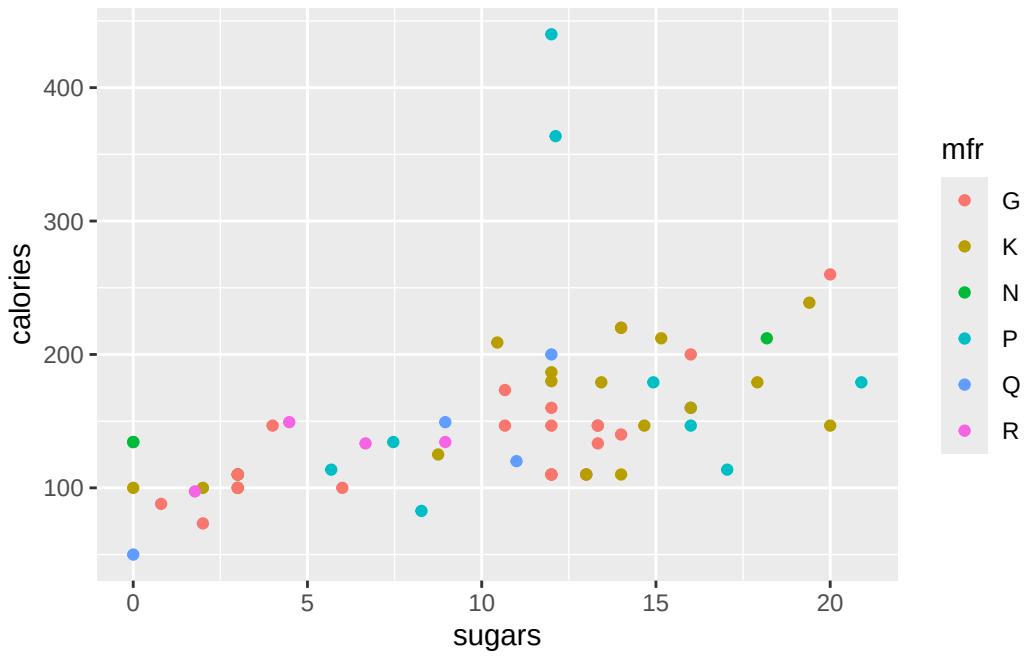
- *Note:* themes are a great way to make your graph look great with minimal work on your end. Check out theme_bw and theme_classic for starters.

You can also add multiple geom_function layers to the same graph

```
p2 <- ggplot(UScereal, aes(x = sugars, y = calories, color = mfr))
```

Overlaying points colored by manufacturer

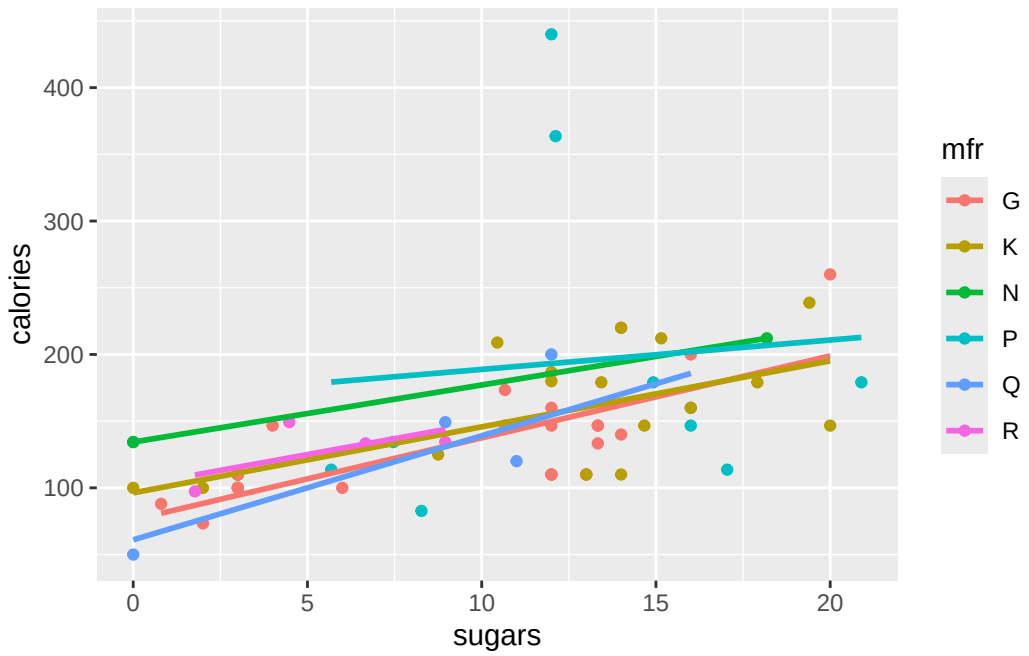
```
p2  + geom_point()
```



Overlaying fitted OLS regression lines by manufacturer

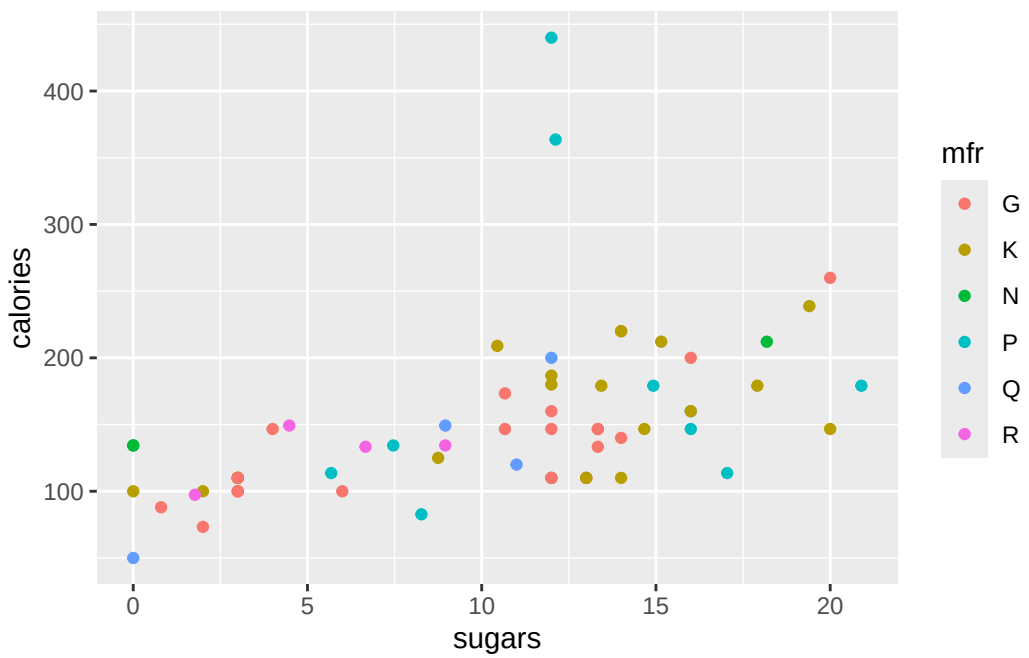
```
p2  +
    geom_point() +
    geom_smooth(method = "lm", se = FALSE)
```

```
`geom_smooth()` using formula = 'y ~ x'
```



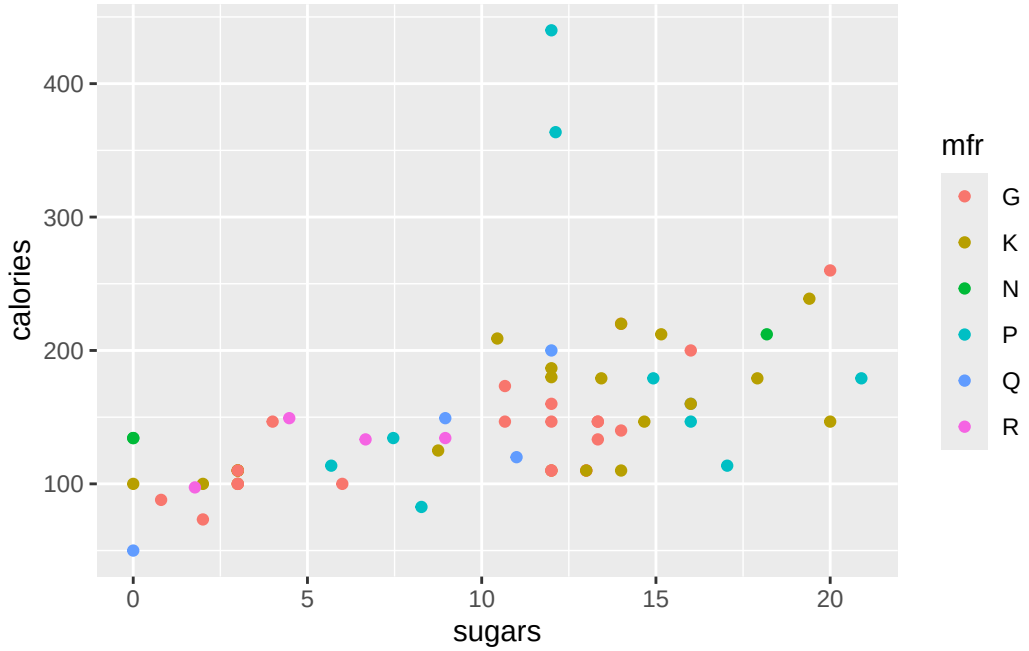
To color by manufacturer - we can put the color in the ggplot function:

```
p2 <- ggplot(UScereal, aes(x = sugars, y = calories, color = mfr))

p2 + geom_point()
```



Or inside the geom_point function:

```
p2 <- ggplot(UScereal, aes(x = sugars, y = calories))

p2 + geom_point(aes(color = mfr))
```

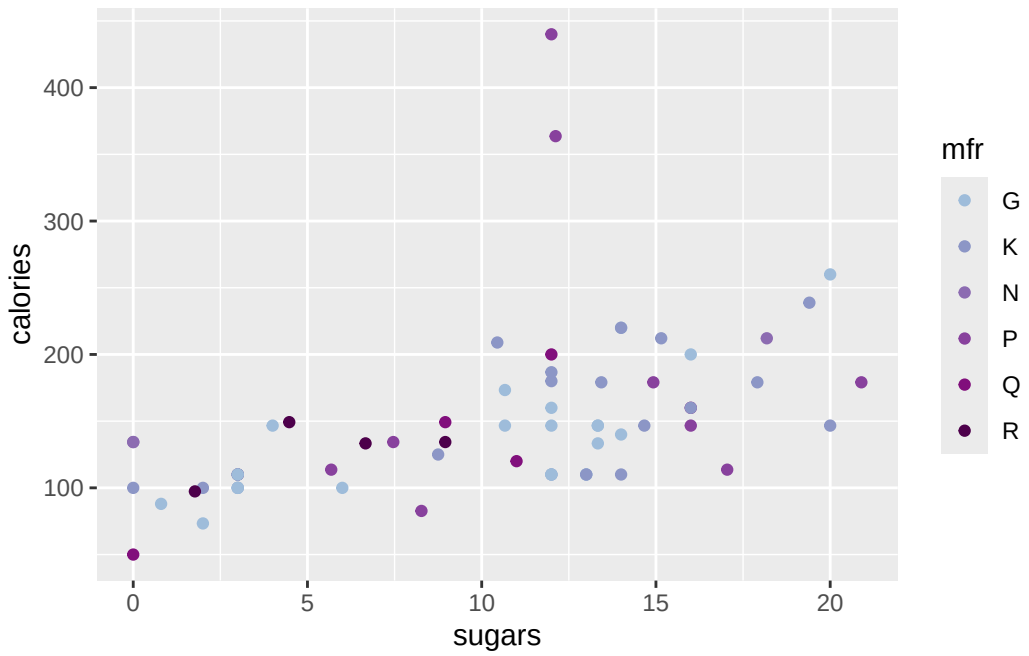


C. Aesthetics: x position, y position, size of elements, shape of elements, color of elements

elements: geometric shapes such as points, lines, line segments, bars and text geometries have their own aesthetics i.e. points have their own shape and size

We can scale colors manually using the appropriately named scale_color_manual function

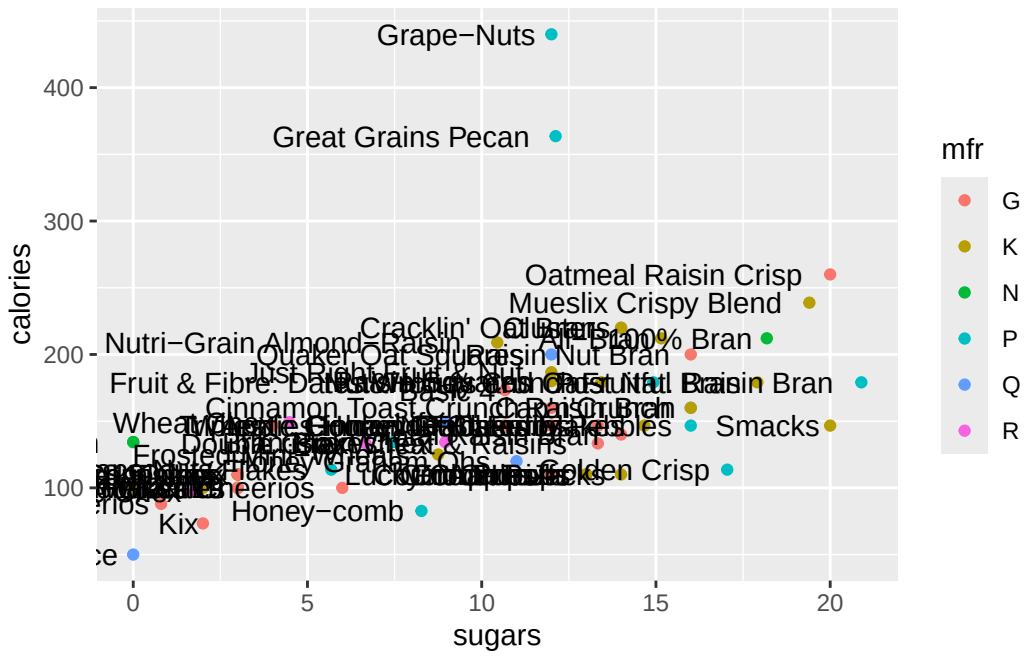
```
my_colors <- c('#9ebcda', '#8c96c6', '#8c6bb1', '#88419d', '#810f7c', '#4d004b')

p2 + geom_point(aes(color = mfr)) +
  scale_color_manual(values = my_colors)
```



Adding Labels to points

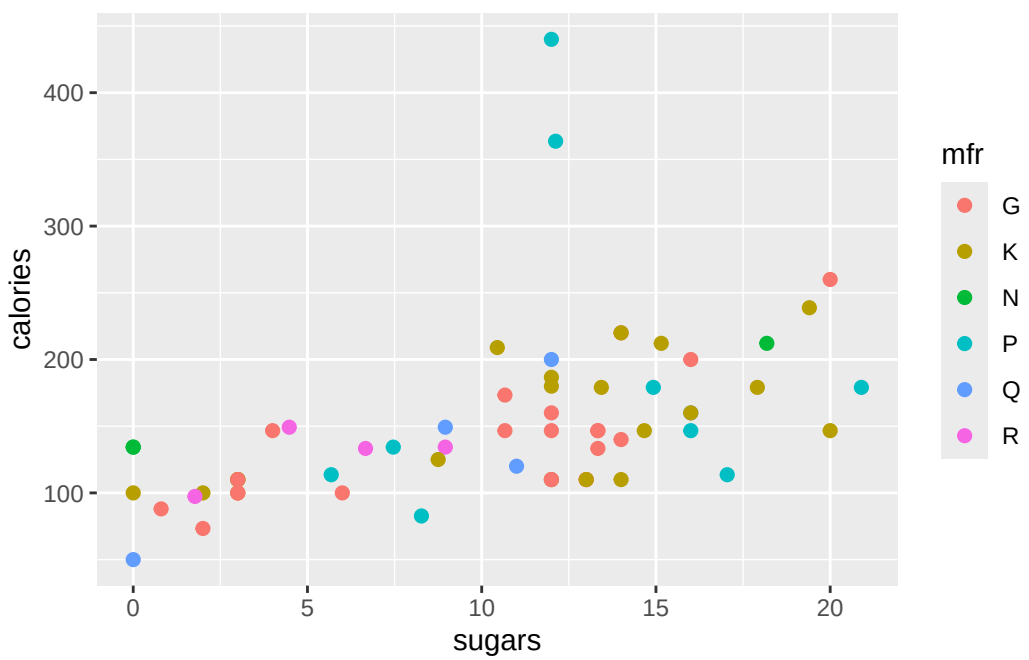
Use the geom_text() layer

```
p2 + geom_point(aes(color = mfr)) +
  geom_text(aes(label = row.names(UScereal)), hjust = 1.1)
```



Changing point size

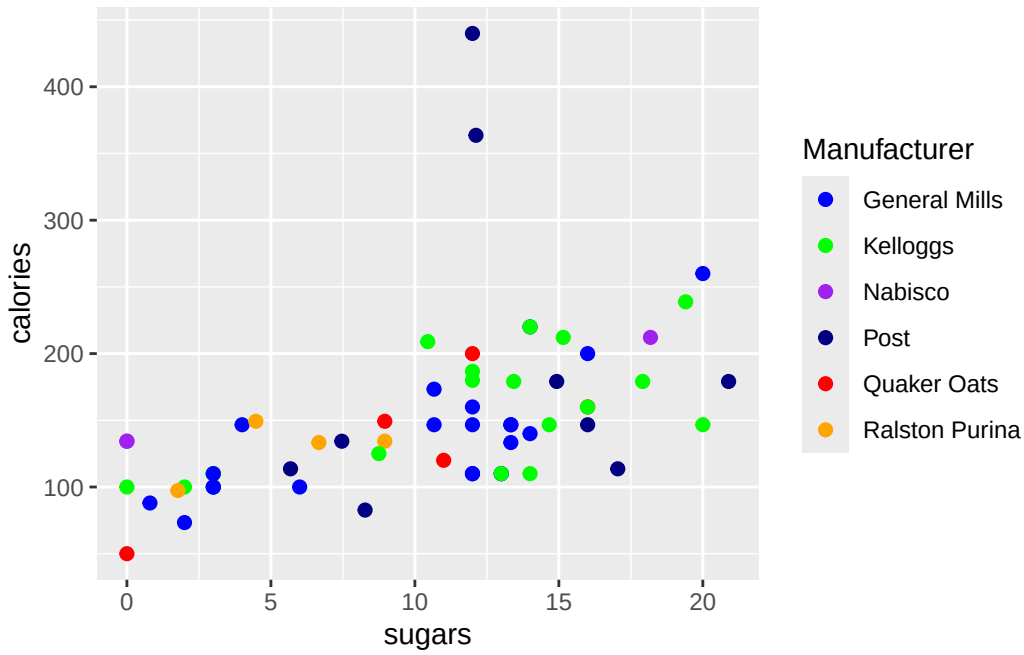
```
p2 + geom_point(aes(color = mfr), size = 2)
```



Editing the legend

Use the scale_color_manual() layer, and the color argument in the labs() layer

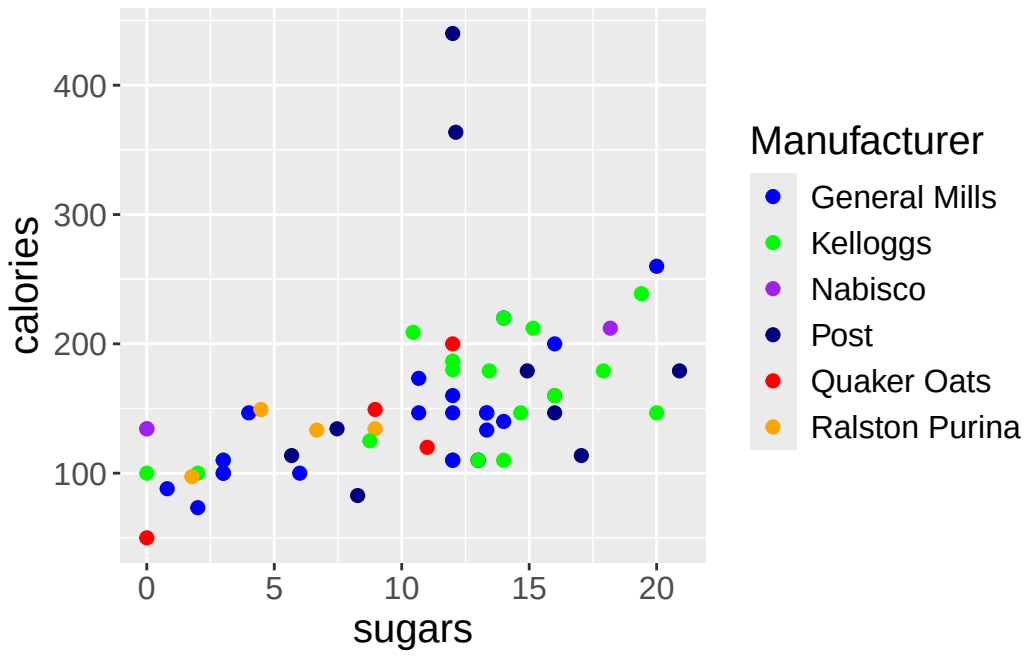
```
p2 + geom_point(aes(color = mfr), size = 2) +
  labs(color = 'Manufacturer') +
  scale_color_manual(values = c('blue', 'green', 'purple', 'navyblue', 'red', 'orange'),
                     labels = c('General Mills', 'Kelloggs', 'Nabisco', 'Post', 'Quaker Oats
```



Changing text size

We can also change the size of legend and axis labels

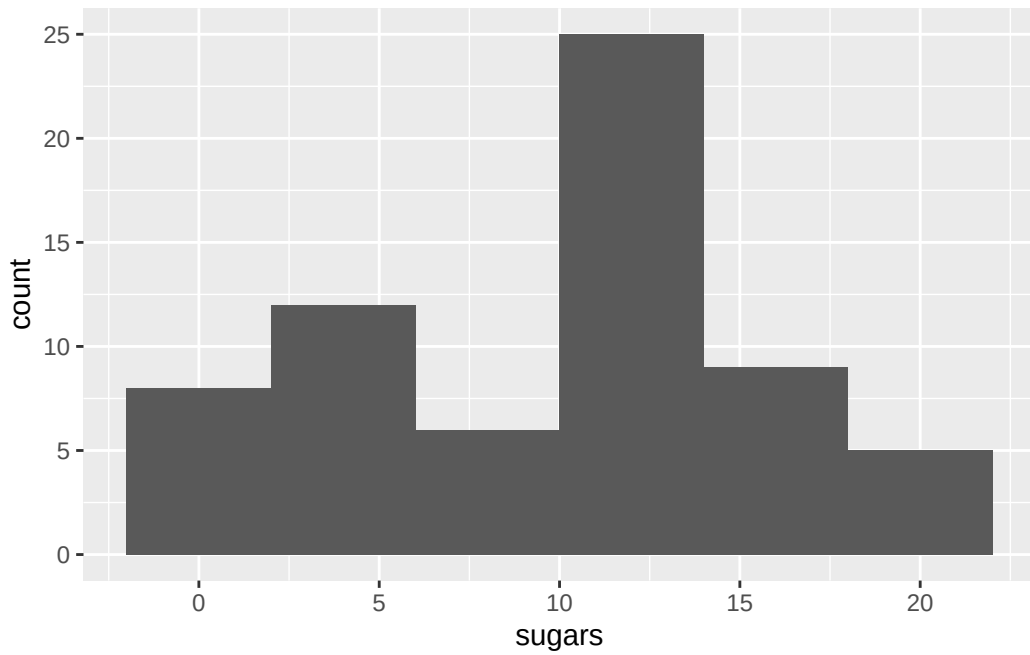
```
p2 + geom_point(aes(color = mfr), size = 2) +
  labs(color = 'Manufacturer') +
  scale_color_manual(values = c('blue', 'green', 'purple', 'navyblue', 'red', 'orange'),
                     labels = c('General Mills', 'Kelloggs', 'Nabisco', 'Post', 'Quaker Oats
  theme(text = element_text(size = 15))
```



D. Faceting - divide a plot into subplots based on the values of one or more discrete variables

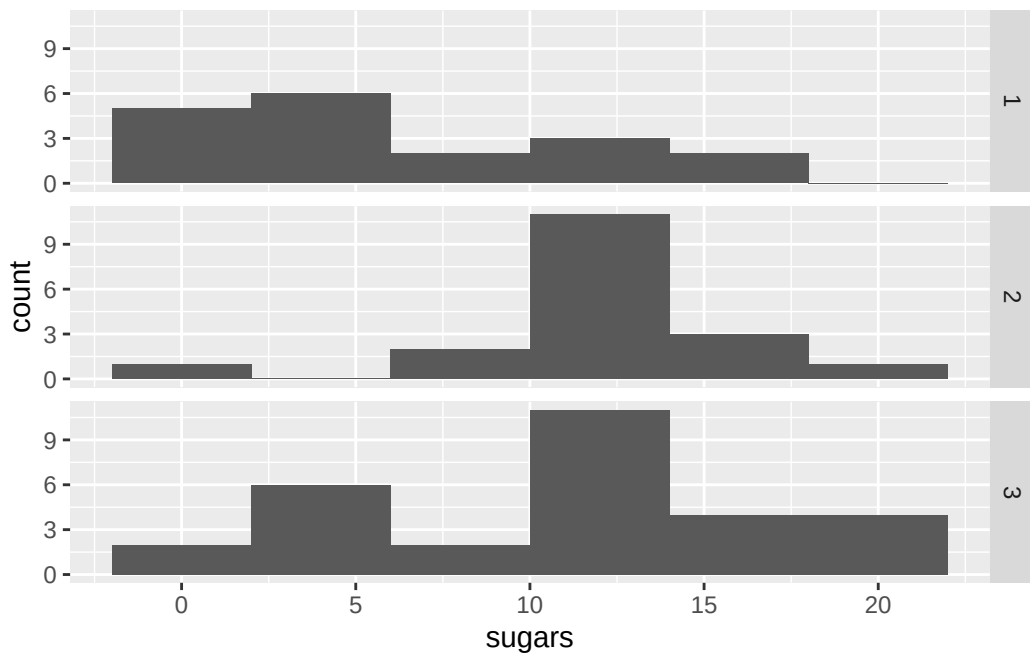
Tip: Use facets to help tell your story Key questions: - How is the distribution of sugar across different shelves? - Are cereals with higher sugar content on lower shelves/at a child's eye level?

```
p3 <- ggplot(UScereal, aes(x = sugars))

p3 + geom_histogram(binwidth = 4)
```



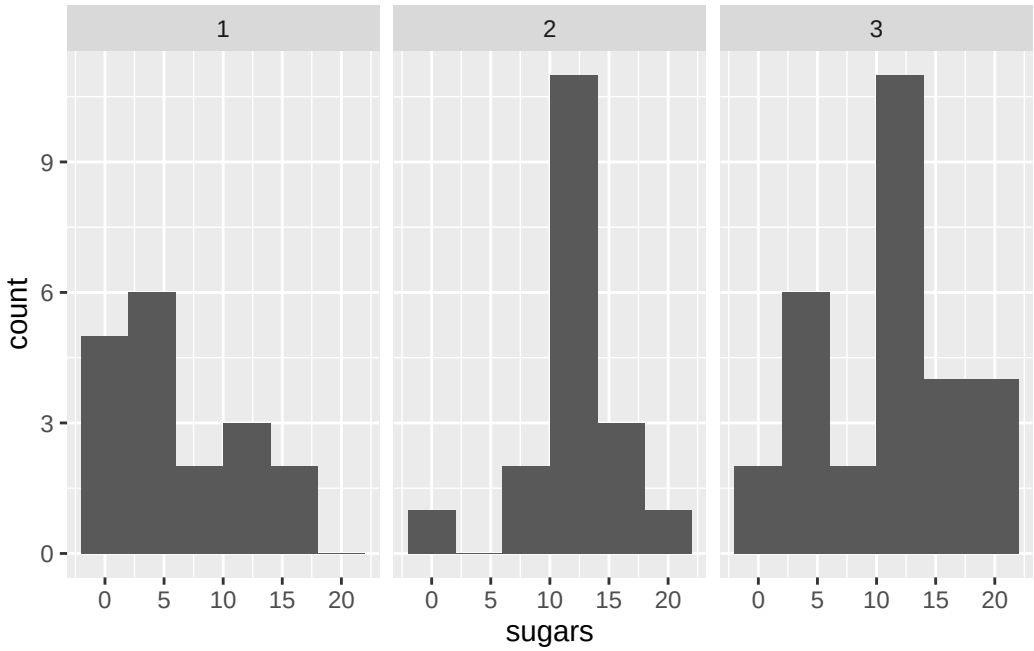
Each graph is in a separate row of the window

```
p3 + geom_histogram(binwidth = 4) +
  facet_grid(shelf ~ .)
```



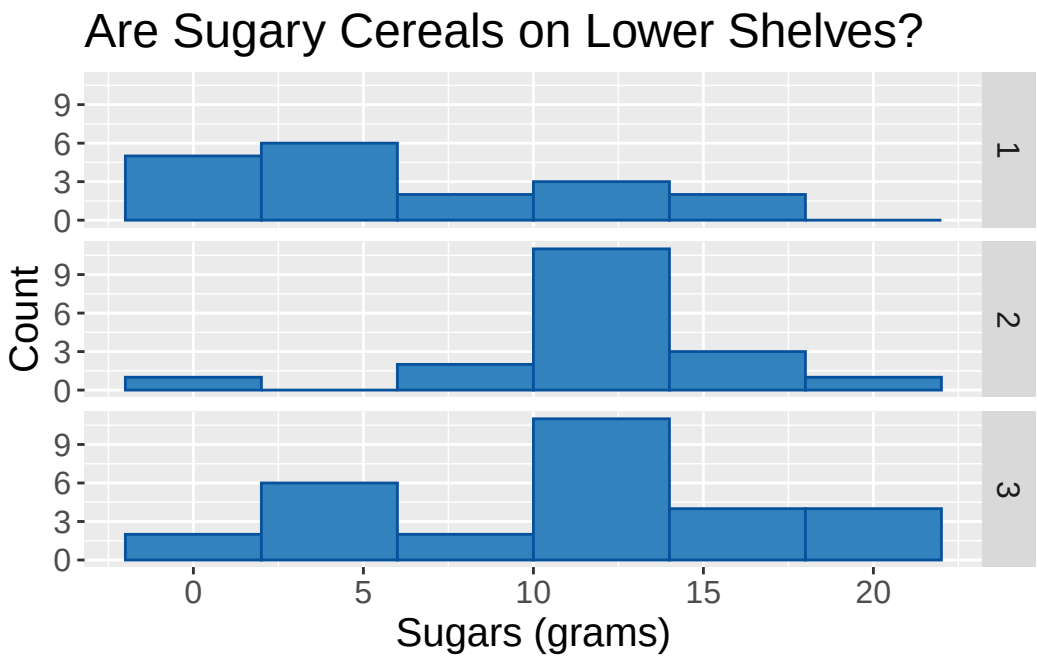
Each graph is in a separate column of the window

```
p3 + geom_histogram(binwidth = 4) +
  facet_grid(. ~ shelf)
```



Finished product

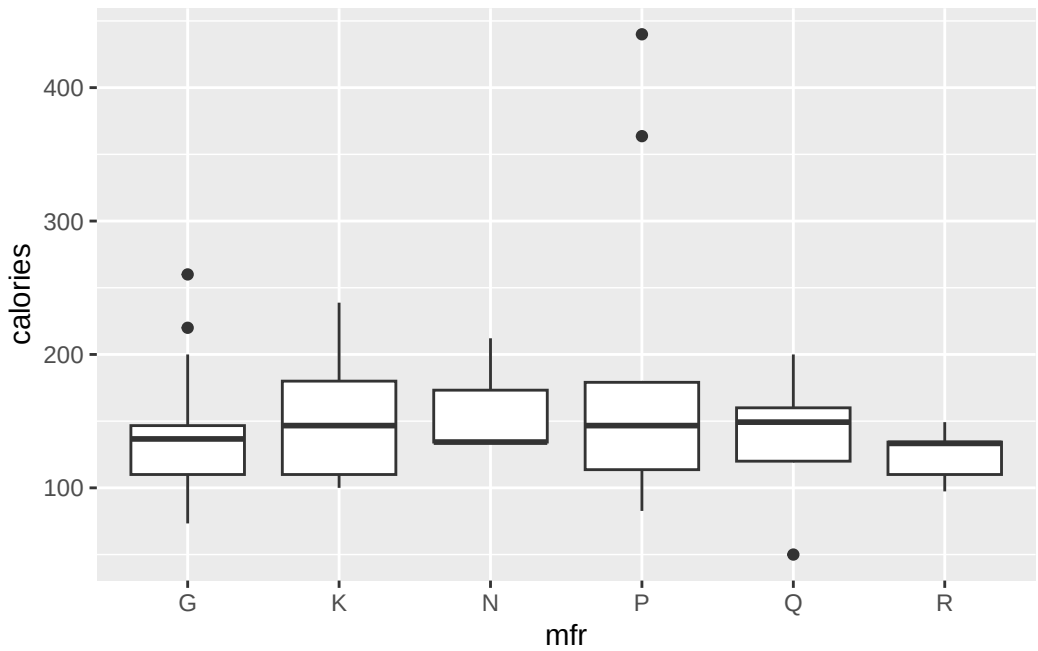
```
p3 + geom_histogram(fill = '#3182bd', color = '#08519c', binwidth = 4) +
  facet_grid(shelf ~ .) +
  theme(text = element_text(size = 15)) +
  labs(title = 'Are Sugary Cereals on Lower Shelves?',
       x = 'Sugars (grams)', y = 'Count')
```



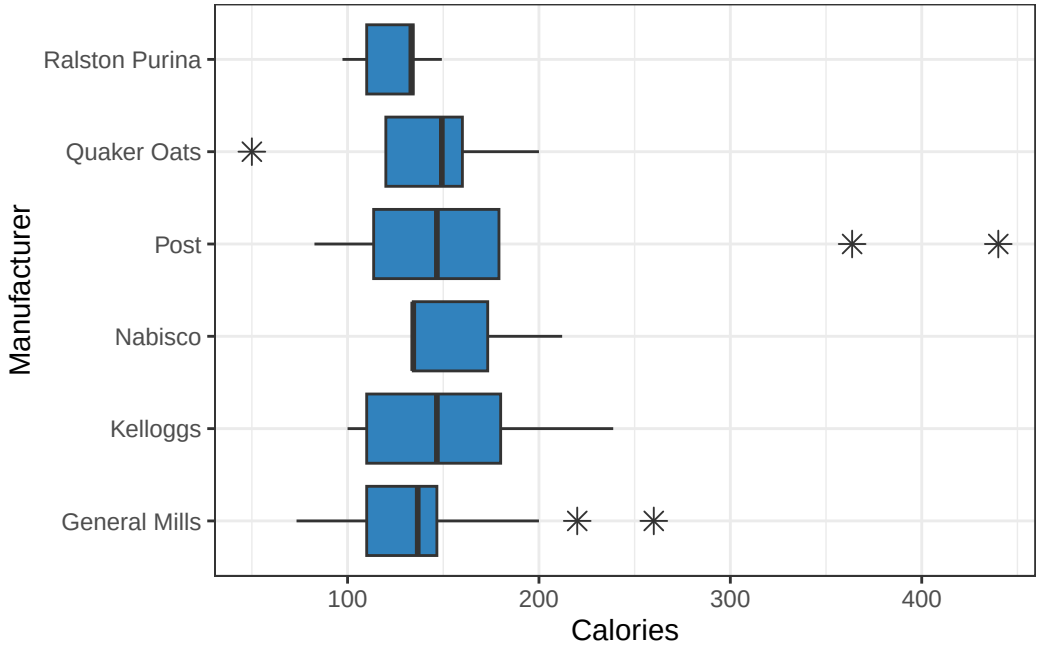
Box Plots

```
p4 <- ggplot(UScereal, aes(mfr, calories))
```

```
p4 + geom_boxplot()
```




```
p4 + geom_boxplot(outlier.shape = 8, outlier.size = 3, fill = '#3182bd') +
  coord_flip() +
  labs(x = 'Manufacturer', y = 'Calories') +
  theme_bw() +
  scale_x_discrete(labels = c('General Mills', 'Kelloggs', 'Nabisco', 'Post', 'Quaker Oats',
```



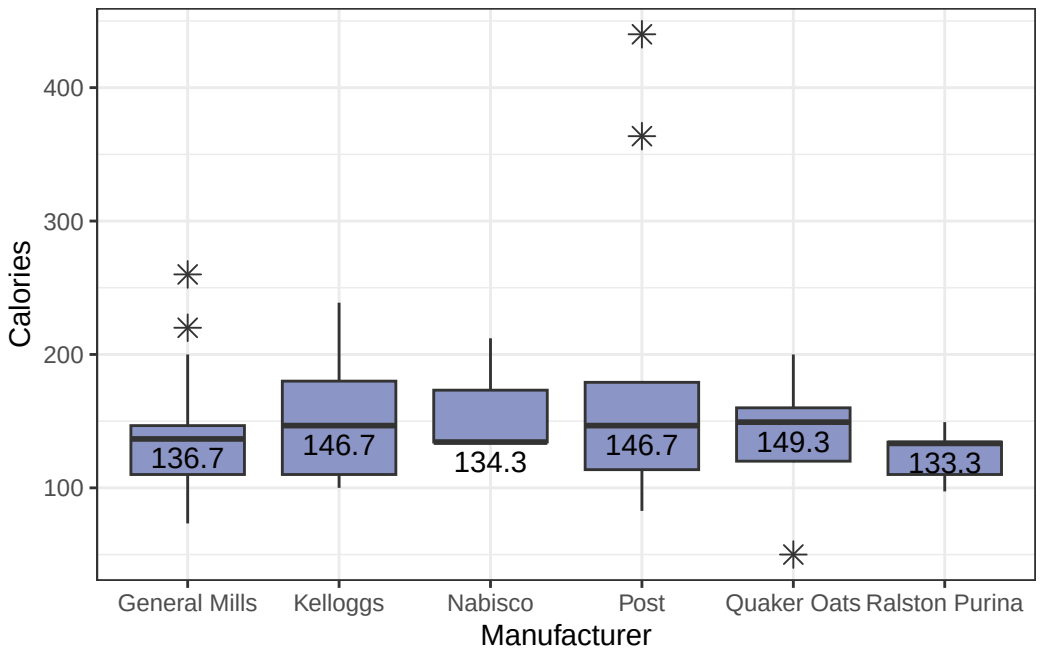
Add median value to boxplot

```
#install.packages('dplyr')
library(dplyr)
```

```
Attaching package: 'dplyr'

The following object is masked from 'package:MASS':

    select

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
p4_meds <- UScereal %>%
  group_by(mfr) %>%
  summarise(med = round(median(calories), 1))
```

```
p4 + geom_boxplot(outlier.shape = 8, outlier.size = 3, fill = '#8c96c6') +
  labs(x = 'Manufacturer', y = 'Calories') +
  theme_bw() +
  scale_x_discrete(labels = c('General Mills', 'Kelloggs', 'Nabisco', 'Post', 'Quaker Oats',
  geom_text(data = p4_meds, aes(x = mfr, y = med, label = med), size = 4, vjust = 1.4)
```

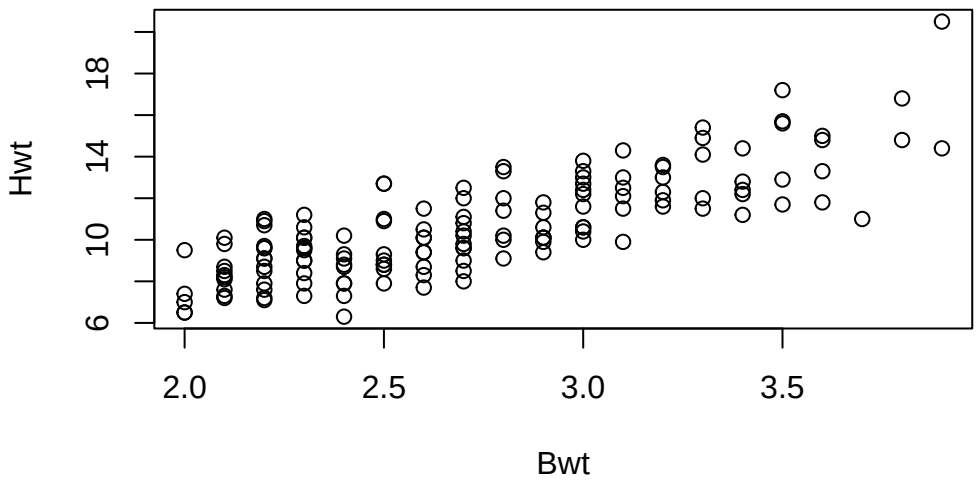


Exercise 2a. Use ggplot2 to improve the graph below:

```
data(cats)
head(cats)
```

```
  Sex Bwt Hwt
1   F 2.0 7.0
2   F 2.0 7.4
3   F 2.0 9.5
4   F 2.1 7.2
5   F 2.1 7.3
6   F 2.1 7.6
```

```
plot(Hwt~Bwt, data = cats)
```

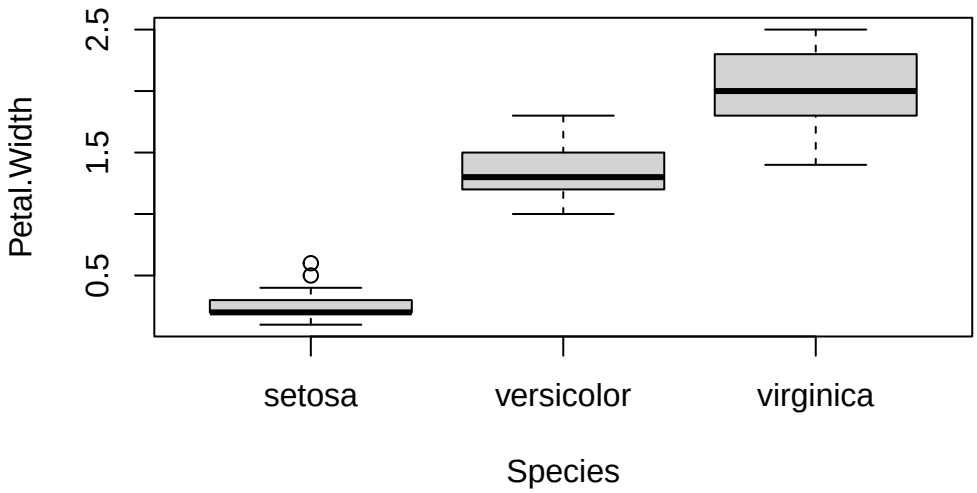


Exercise 2b. Use ggplot2 to improve the graph below:

```
data(iris)
head(iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```

```
boxplot(Petal.Width ~ Species, data = iris)
```



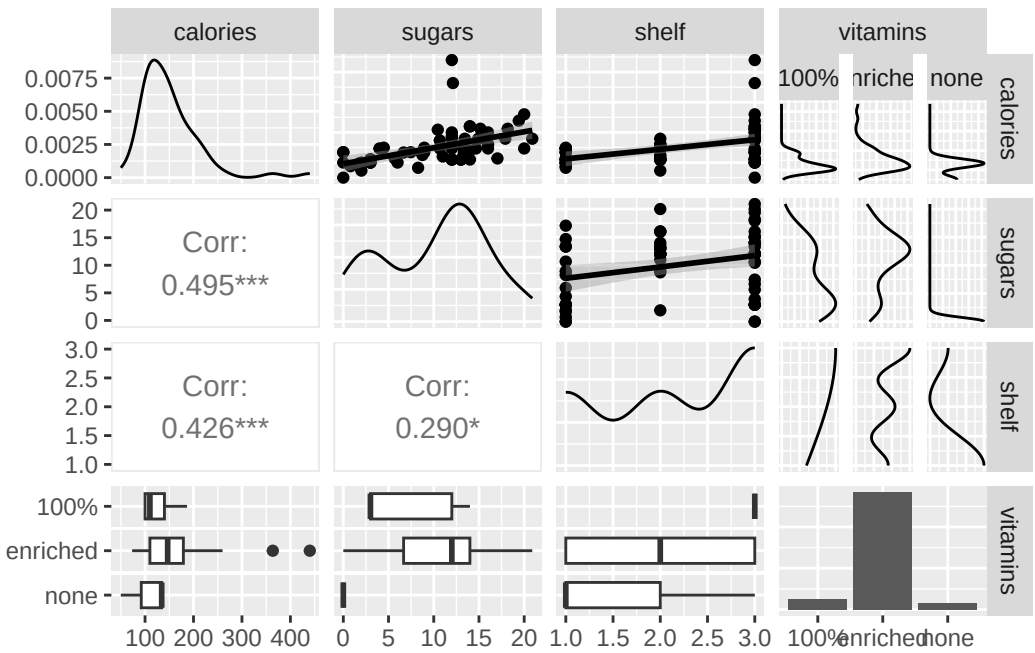
III. Other Graphics

Scatterplot Matrix

```
#install.packages('GGally')
library(GGally)
```

```
Registered S3 method overwritten by 'GGally':
  method from
  +.gg   ggplot2
```

```
ggpairs(UScereal[, c(2, 8, 9, 11)],
        upper = list(continuous = 'smooth', combo = 'facetdensity', discrete = 'blank') ,
        lower = list(continuous = 'cor', combo = 'box'))
```



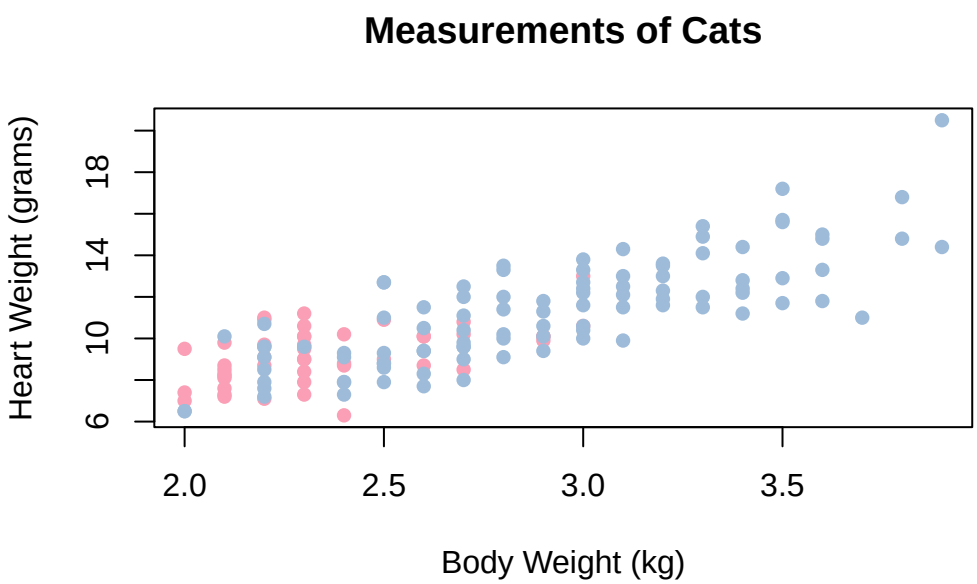
Maps http://bcb.dfci.harvard.edu/~aedin/courses/R/CDC/maps.html http://rstudio.github.io/leaflet/maps, choroplethr,

IV. Exercise Solutions

Below are just examples of cleaned up graphics. There are many solutions on how to improve these graphs

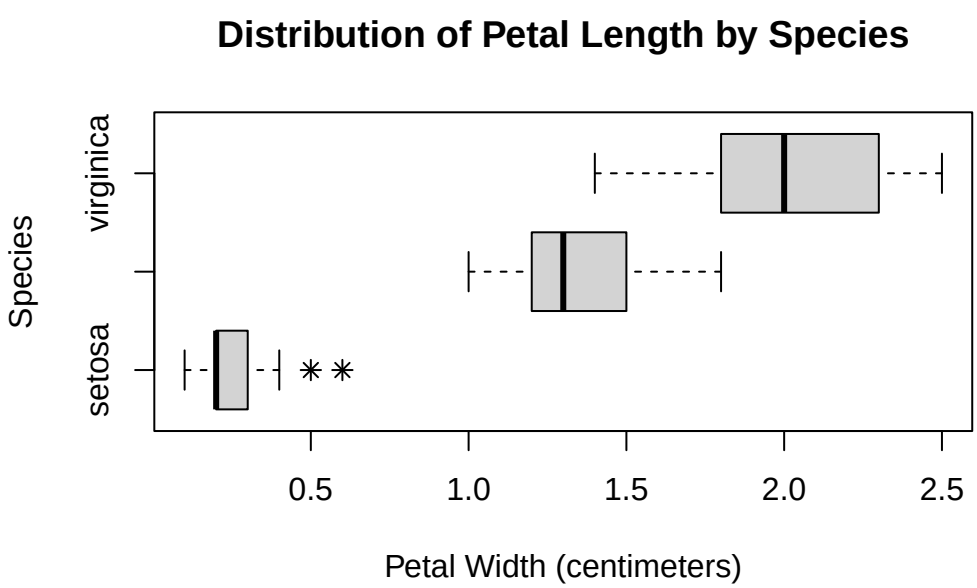
Exercise 1a

```
palette(c('#fa9fb5', '#9ebcda'))
plot(Hwt~Bwt, data = cats, ylab  = 'Heart Weight (grams)', xlab = 'Body Weight (kg)', main =
```



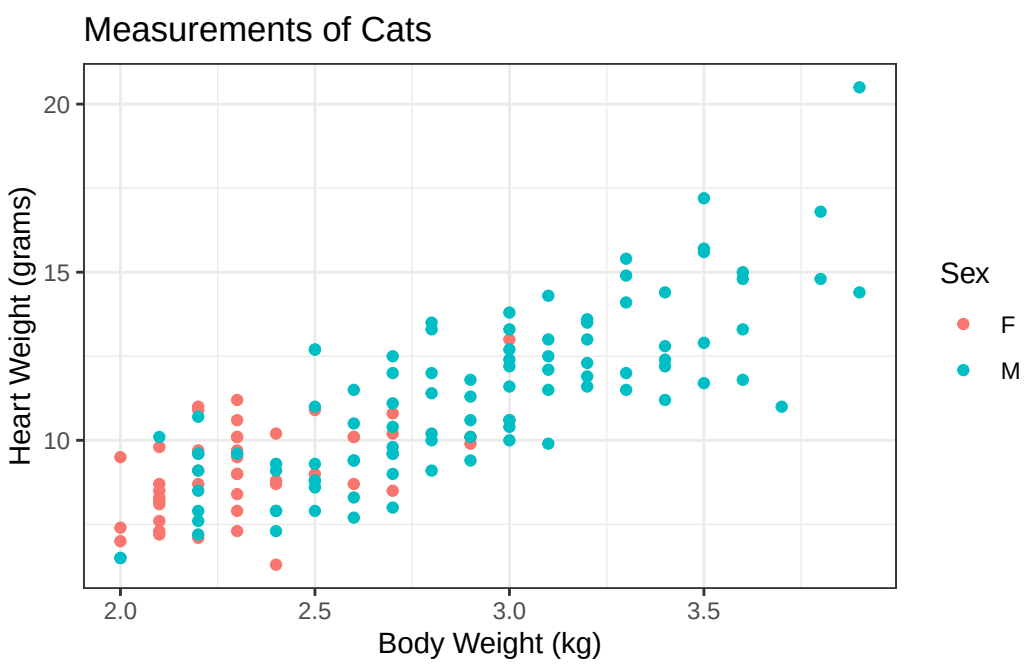
Exercise 1b

```
boxplot(Petal.Width ~ Species, data = iris, xlab = 'Petal Width (centimeters)',
        main = 'Distribution of Petal Length by Species', pch = 8, horizontal = TRUE, col =
```



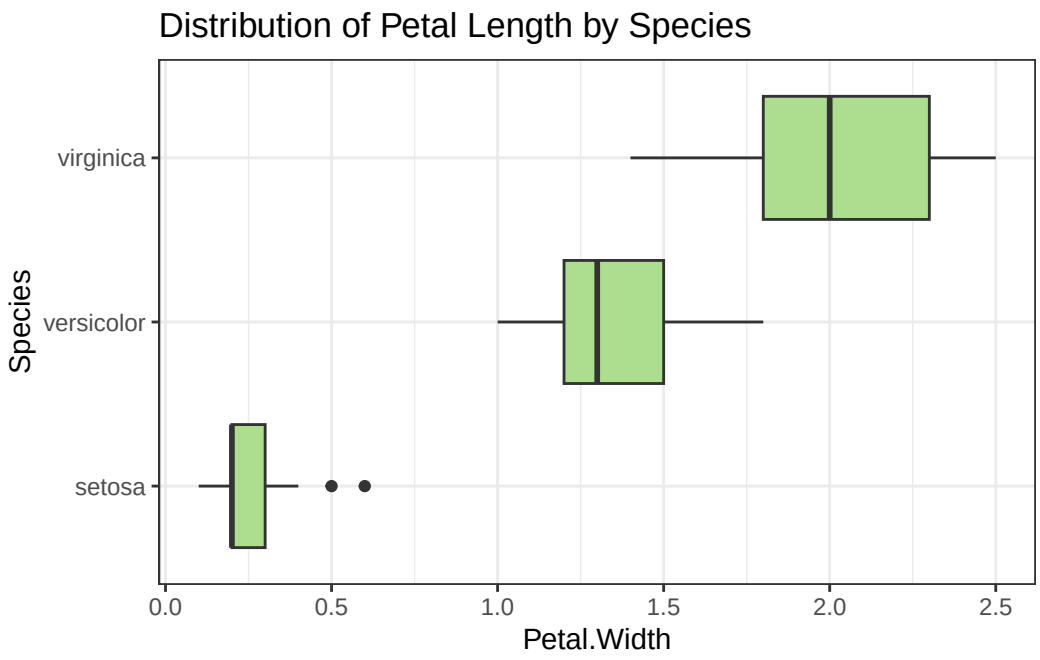
Exercise 2a

```
e2a <- ggplot(cats, aes(Bwt, Hwt, color = Sex))
e2a + geom_point() +
  theme_bw() +
  labs(title = 'Measurements of Cats', x = 'Body Weight (kg)', y = 'Heart Weight (grams)')
```



Exercise 2b

```
e2b <- ggplot(iris, aes(Species, Petal.Width))
e2b + geom_boxplot(fill = '#addd8e') +
  theme_bw() +
  coord_flip() +
  ggtitle('Distribution of Petal Length by Species' )
```


Distribution of Petal Length by Species
virginica
versicolor
setosa
Species
0.0
0.5
1.0
1.5
2.0
2.5
Petal.Width